

Intertwining Logistic Map and Cellular Automata Based Color Image Encryption Model

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Abstract— In this paper a color image encryption involving pixel confusion and diffusion governed by three different series generated by intertwining logistic map is proposed. This symmetric encryption model completes in two steps. First, the original image pixels are xor-ed with randomly generated data having values between 0 and 255. The xor-ed data is permuted by a shuffling engine. Secondly, diffusion of the permuted data is achieved by the help of Elementary Cellular Automata based diffusion mechanism. The decoding of the original information is not possible in the absence of knowledge of appropriate rules of the cellular automata and their order of use. The experimental results demonstrated outstanding performance for various tests conducted. Brute force resistance, larger key space, and good resistance against statistical and differential attacks comes out to be key features of the algorithm.

Keywords— *Elementary Cellular Automata; Pixel Shuffling; Color Image Encryption; Intertwining Logistic Map; Diffusion*

I. INTRODUCTION

Increasing digital dependency raised higher expectations from information security domain. More secure, robust and trusted platforms are emerging in the recent time. These systems are backed by several encryption algorithms. A number of image encryption methods have been proposed in literature [1-2]. Bulk data capacity, high correlation among adjacent pixels, and very low sensitivity limits the use of many data encryption algorithms. Other reasons for not using them are speed, data type, computational power and high complexity [3]. Randomness, sensitivity to initial conditions [4], control parameters, good confusion and relatively fast implementation of chaos based systems made these systems suitable for image encryption.

Many encryption systems have been developed in the recent past utilizing chaotic systems such as Logistic map, Cat map [5], Chen's system, Lorenz's system, 2-D cellular automata [6] and many more. In general, the encryption key in these symmetric encryption methods is deduced by iterating one of the used chaotic systems. Further, other keys or sub keys are deduced by again using same systems. Many advantages and limitations of various systems are studied in literature. Many of the proposed algorithms have used Logistic map for image encryption [7]. The unwanted properties of logistic map such as blank window, stable window and uneven distribution of

sequence attracted many researches towards the intertwining Logistic Map which overcome these limitations.

In this paper, an encryption algorithm based on a chaotic map and cellular automata is proposed. The algorithm uses both permutation and diffusion for enhanced security. Further, a chaos based random number generator is used such that the encryption process becomes sensitive to initial condition of the chaotic system.

Here in this work section: II gives brief introduction to cellular automata. Section: III and IV describes the key generation and proposed pixel shuffling procedure. Encryption, diffusion and corresponding decryption algorithm are described in section: V, VI and VII respectively. Different security tests were performed and their corresponding results are discussed in detail in section: VIII. In section: IX conclusions are made based on the entire proposed algorithm and corresponding test results.

II. ELEMENTARY CELLULAR AUTOMATA

Von. Neumann proposed cellular automata (CA) theory of self-replication for different robots. It was found that the cellular automata exhibit dynamical behavior with some discrete parameters like time and space. The behavior of a CA cell is governed by some set of conditions known as the rules. A rule is simply a logical condition which derives the next state of a particular cell by considering the neighborhood of the current cell. There are a no. of ways in which a neighborhood of a cell is defined in literature. Some of the ways to define the neighborhood are 3-neighborhood, Von-Neumann and Moore's neighborhood. Mathematically, cellular automata can be understood by the following set of equations.

$$CA_i^{t+1} = f(C_{i-r}^t, C_{i-r+1}^t, \dots, C_i^t, C_{i+1}^t, \dots, C_{i+r}^t)$$

Where, C_i^t is i^{th} cell at time t , and C_i^{t+1} is i^{th} cell at time $t+1$ and r is neighborhood radius. In the proposed work, linear Cellular Automata is used where $r=1$ making 3 cells at a time for consideration (see figure: 1).

The evolution of the cellular automata generates random numbers which can be used in the encryption systems. The

random number generation is governed by the neighbor cells, initial state and rule used. The evolution of CA changes the initial state dynamically.

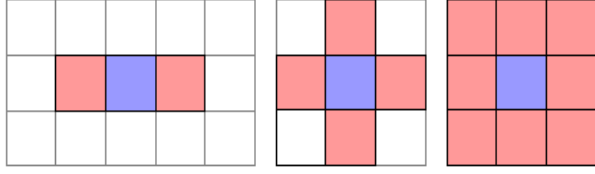


Fig. 1. Elementary Cellular Automata neighborhood in which central cell is cell in consideration and different types of neighborhood is represented as colored cells.

III. KEY GENERATION

A randomly generated binary key K of length 128 bits is used in algorithm. The key is used in all the processes of encryption algorithm such as 3-D matrix generation, shuffling of pixels, and diffusion process. From this key K , 16 sub-keys are generated (each consisting of 8 bits) which are used to generate different parameters for encryption process. Here each sub-key can have a decimal equivalent value between 0 and 255. The sub keys can be generated as per following relation.

$$K_i = Key(8 \times i + 1) \text{ to } Key(8 \times (i + 1))$$

where, $i = 0$ to 15. After obtaining sub-keys different input parameters of shuffling and encryption algorithm are generated. Initial conditions of Logistic map used in various steps can be computed as

$$x_1 = (k_1 \times k_5) / (k_4 \times k_{16}) \text{ Mod } 1$$

$$y_1 = (k_3 \times k_9) / (k_{10} \times k_{15}) \text{ Mod } 1$$

$$z_1 = (k_7 \times k_{11}) / (k_6 \times k_{12}) \text{ Mod } 1$$

$$x_2 = (k_{10} \times k_3) / (k_1 \times k_2) \text{ Mod } 1$$

$$y_2 = (k_{13} \times k_7) / (k_{12} \times k_5) \text{ Mod } 1$$

$$z_2 = (k_4 \times k_{10}) / (k_{15} \times k_8) \text{ Mod } 1$$

Different scaling factors for diffusion and other processes can be computed as

$$s_1 = (k_{13} / k_9) \times (k_{16} \times k_{14}) + 113$$

$$s_2 = (k_{10} / k_7) \times (k_{11} \times k_{15}) + 117$$

$$sf_1 = (k_{14} / k_4) \times (k_2 \times k_{16}) + 131$$

$$sf_2 = (k_3 / k_5) \times (k_1 \times k_{14}) + 119$$

IV. SHUFFLING

Shuffling process is used to permute the input image so as to avoid revealing any statistical information. For input image pixels, the shuffling process was carried out with constantly updating input pairs (x_1, x_2) , (y_1, y_2) and (z_1, z_2) respectively and keeping (s_1) and (s_2) as fixed. The shuffling of each input linear array is achieved 5 times. At each iteration the last values obtained becomes the initial parameters for the next iteration. Each iteration generates a number between 0 and 1 which is first scaled with the scaling factor s_1 or s_2 .

The Logistic Map used is:

$$q = 4 \times in(1 - in) \quad (1)$$

Further, the resulting values are scaled up as:

$$Array = \lfloor q \times scalingFactor \rfloor 4 \text{ Mod } (lc) + 1 \quad (2)$$

Here, in is one of the initial values for Logistic Map, q is output of Logistic Map, scalingFactor may be either s_1 or s_2 . Therefore, the shuffling of an array (say Q for instance) involve the following steps:

Step 1 As the length of input array Q is lc i.e. 256, two arrays (array Q_1 and array Q_2) consisting of random numbers between 1 and 256 (both including) are generated using relation:

$$Array = \lfloor q \times scalingFactor \rfloor 4 \text{ Mod } (256) + 1 \quad (3)$$

Step 2 Take the i^{th} value of Q_1 say $Q_1(i)$ and using it as index, find the value $Q(Q_1(i))$

Step 3 Substitute $Q(Q_1(i))$ with $Q(Q_2(i))$. Therefore $Q(Q_1(i)) \rightarrow Q(Q_2(i))$ Hence the elements of Q get shuffled as $Q(Q_1(i)) \rightarrow Q(Q_2(i)) \forall i$.

Step 4 Steps 2 and 3 are repeated cl times which in the end results into a shuffled array Q .

Step 5 Repeat Step 1 to 4, five times to get the final shuffled array Q' .

V. ENCRYPTION PROCESS

The process completes in following steps.

(i) First arrange the image as a linear array of size $row \times column \times 3$.

(ii) Iterate the Logistic map with three random initial states 30 (or more) times and obtain s .

(iii) Using 30th iteration result of s iterate Intertwining Logistic map $row \times column \times 3$ row times and obtain s_1 , similarly using second last iterations of the Logistic map obtain s_2 .

Mathematical expression of intertwining logistic map used [8]:-

$$\begin{aligned}
x_{n+1} &= \left[\mu \times k_1 \times y_n \times (1 - x_n) + z_n \right] \text{Mod} 1 \\
y_{n+1} &= \left[\mu \times k_2 \times y_n + z_n \times (1 + x_{n+1}^2) \right] \text{Mod} 1 \\
z_{n+1} &= \left[\mu \times (y_{n+1} + x_{n+1} + k_3) \times \sin(z_n) \right] \text{Mod} 1
\end{aligned}$$

Where μ have values between 0 and 3.9999, $k_1 > 33.5$, $k_2 > 37.9$, $k_3 > 35.7$.

(iv) Add all the columns of s_2 and multiply the result with a scaling factor sf_1 . Obtain absolute values and convert to integer values using $\lceil \cdot \rceil$ (ceiling function) and take MOD 256.

(v) Bitxor result of step (iv) with the pixels of the plain image which result into noisy image I_1 .

(vi) Shuffle I_1 using the shuffling mechanism explained in previous section with input parameters deduced from s_1 and get I_2 .

(vii) The pixels in I_2 were diffused using the “Diffusion Mechanism” which is explained in the next section (VI).

In the encryption process explained above, shuffling permutes the pixels of the image which results to loss of correlation of the pixels and diffusion escalates a minor change in the pixel to subsequent pixels and gives final encrypted image.

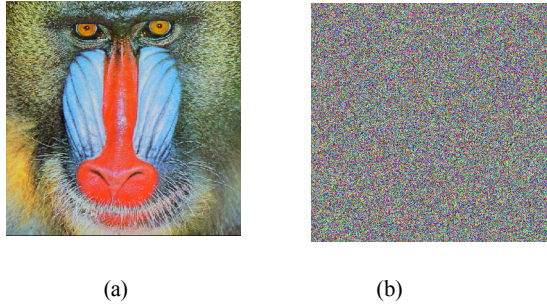


Fig.2. Result of encryption process, a: Original image, b: cipher image of a using proposed scheme

VI. DIFFUSION MECHANISM

Various diffusion mechanisms propagate the change in pixel to the subsequent pixels. The mechanism used is very simple yet effective to provide better security to the image. All the steps involved are explained below.

- (i) Using s_1 generate f_1 by multiplying s_1 with a scaling factor sf_2 .
- (ii) Generate sequences ru and $iter$ using f_1 by taking MOD with 256 and 14 respectively.
- (iii) Declare a variable i to 1 and iterate it to the length of I_1 following below given equation

First, set $sw_i = 254$;

$$\begin{aligned}
I_v &= \left\lceil sw_i + \left| (s_i, 1) \times (s_i, 3) \times 10^3 \right| \times adj \right\rceil \\
I_3 &= (I_{2i} + I_v) \text{Mod} 256 \oplus iter_i
\end{aligned}$$

Where adj is obtained using cellular automata with rule ru_i and data as $\lceil |f_{1i}| \rceil \text{Mod} 256$. After each iteration update sw_i to output of the previous step.

VII. DECRYPTION PROCESS

The decryption process involves the steps described for encryption in reverse order. At first, the checksum of the cipher image is generated and is compared with the checksum padded by the sender with the image, if both are same then cipher image is decrypted otherwise cipher is discarded as tempered and receiver asks sender for retransmission of cipher image. For decryption cipher image pixels are diffused again with the given relationship.

$$\begin{aligned}
I_u &= \left\lceil sw_i + \left| (s_i, 1) \times (s_i, 3) \times 10^3 \right| \times adj \right\rceil \\
I_3^{(r)} &= (I_{3i} \oplus iter_i + 256 - I_u) \text{Mod} 256
\end{aligned}$$

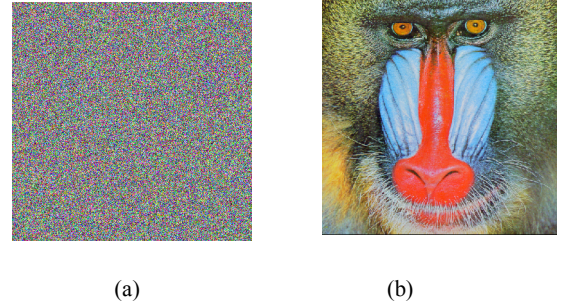


Fig.3. Result of decryption process, (a) Received cipher image, (b) Original image retrieved after decryption process.

VIII. EXPERIMENTAL RESULTS AND ANALYSIS

High key sensitivity, differential attack resistance, high NPCR score and brute force resistance are key requirements of an image encryption algorithm [5]. The proposed algorithm was investigated for above mentioned tests.

(i) Dependence on key: This is a required feature of an image encryption algorithm. It is computed as the difference of two cipher images of a single source image after encrypting with very minute (often only one bit) change in key. The results clearly validates the proposed algorithm, where two cipher images were found completely non identical.

(ii) Key Space: The proposed algorithm uses a key of 128 bits length to generate different parameters of chaotic systems and other processes. As it would have 2^{128} different possible key

combinations which are large enough to resist any kind of brute force attack.

(iii) **Histogram analysis:** Figure 4(a-f) shows uniform distribution of pixel in cipher image using the proposed algorithm. The uniform distribution reduces the chance of reconstruction of image from the cipher generated.

The histogram shows that the pixels are randomly distributed in the entire image. The difference in histograms of images can be seen in Figure: 4(a-f).

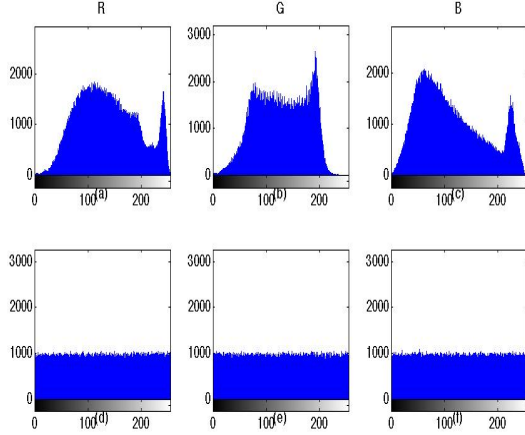


Fig.4. Histogram of images, (a,b,c) histogram of original plain text image (RGB color), (d,e,f) histogram of first cipher image. The histogram clearly indicates the distribution of pixels in entire image.

(iv) **Correlation:** The test investigates the extent to how much two adjacent pixels are correlated to each other. The value of correlation coefficient lies between -1 to 1. A correlation value of 1 indicates high positive correlation and -1 indicates highly negative correlation. For a good encryption algorithm, correlation values are expected to be as close to zero. Figure: 5 (a-f) shows the correlation between 5000 randomly chosen pixels. The correlation is calculated as following.

$$d_a = \sqrt{(N \sum_{n=1}^N x_n^2 - (\sum_{n=1}^N x_n)^2)(N \sum_{n=1}^N y_n^2 - (\sum_{n=1}^N y_n)^2)}$$

$$r_{x,y} = \frac{N \sum_{n=1}^N (x_n y_n) - \sum_{n=1}^N x_n \sum_{n=1}^N y_n}{d_a}$$

Here N is total number of pixels selected for test and x_n, y_n are two adjacent pixels. Correlation of plain text image, image obtained after XOR process and final encrypted images are tabulated below.

TABLE I
Correlation of adjacent pixels in R,G and B channels.

Plane	Stage	Horizontal	Vertical	Diagonal
RED	Original	0.8660	0.9231	0.85434
	Cipher	-0.0019	0.0018	-0.0018
GREEN	Original	0.7650	0.8655	0.7348
	Cipher	-0.0014	0.0007	-0.0034
BLUE	Original	0.8809	0.9073	0.8399
	Cipher	-0.0009	-0.0043	0.0002

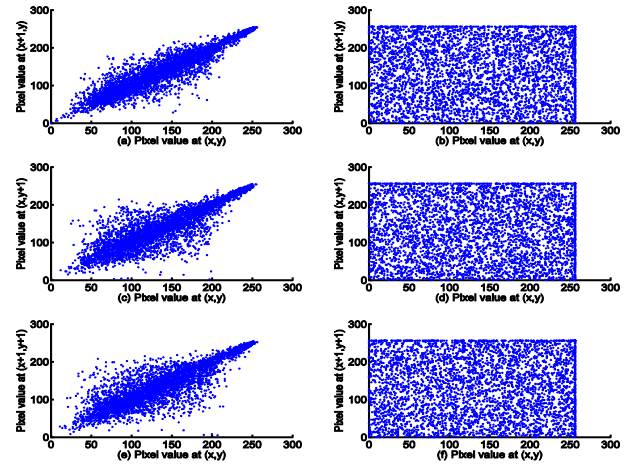


Fig.5. Correlation analysis of the encryption algorithm, (a-b) Horizontal correlation of plain text and cipher image, (c-d) Vertical correlation of plain text and cipher image, (e-f) Diagonal correlation of plain text and cipher image.

TABLE II
NPCR and UACI values of different images

Image	Size	NPCR %	UACI %
Baboon	512 x 512	99.613571	33.45746
Lena	512 x 512	99.618275	33.46680
Tiffany	512 x 512	99.606831	33.46554
Pepper	512 x 512	99.603907	33.45038
Drop	512 x 512	99.607594	33.48086

(v) **Differential Attack:** It is the process of finding out any relationship between two cipher images generated from a single plain text image with one pixel value difference. If the resulting cipher images are entirely different with NPCR (Number of Pixel Change Rate) greater than 99.6%, the cipher images are considered attack proof, otherwise chances of retrieving some useful information are there [9]. In order to test this, two identical plain text images with only one pixel value difference were taken and encrypted with the proposed algorithm. Here two images I_1 and I_2 with gray-scale pixel values $I_1(j, k)$ and $I_2(j, k)$ respectively, a third array $Z(j, k)$ is obtained as: if $I_1(j, k) = I_2(j, k)$ then $Z(j, k) = 1$, otherwise $Z(j, k) = 0$.

The NPCR is calculated as:

$$NPCR = \left\{ \frac{\sum_{j,k} (Z(j,k))}{M \times N} \right\} \times 100\% \quad (4)$$

while the average amount of change or average intensity of difference is calculated as the UACI (Unified Average Changing Intensity). here UACI is calculated as:

$$UACI = \frac{\sum_{j,k} |I_1(j,k) - I_2(j,k)|}{255 \times M \times N} \times 100\% \quad (5)$$

On testing the NPCR and UACI values for different images of same and different sizes in all the intermediate steps of encryption following results are obtained (tabulated). Following table compares the NPCR and UACI values of different cryptosystem (Gray images).

TABLE III
NPCR and UACI comparison with different models

Test	[10]	[11]	[12]	[13]	[14]	Proposed
NPCR%	99.62	99.58	99.67	99.61	99.60	99.61
UACI%	33.42	33.42	27.88	33.41	33.46	33.47

The below table compares correlation values of different cryptosystem for gray images.

TABLE IV
Correlation comparison of different algorithms with the proposed algorithm.

Dimension	[10]	[11]	[12]	[13]	Proposed
Horizontal	-0.0019	0.0172	0.0107	-0.0209	-0.0013
Vertical	-0.0028	0.0099	0.0141	-0.0144	-0.0010
Diagonal	0.0008	0.0330	0.0097	-0.0350	0.0004

(vi) **Information Entropy:** As per Shannon's theory the information entropy H_S of a given plain text symbol S_i is calculated as per the following formula:

$$H_s = \sum P_{s_i} \left(\log \frac{1}{P_{s_i}} \right)$$

here P_{S_i} is probability of symbol S_i . The value of H_S should ideally be 8, as if one transmits 108 characters/ symbols with equal probability. The calculated value of H_S is 7.8686 for cipher image which is close to theoretical value of 8.

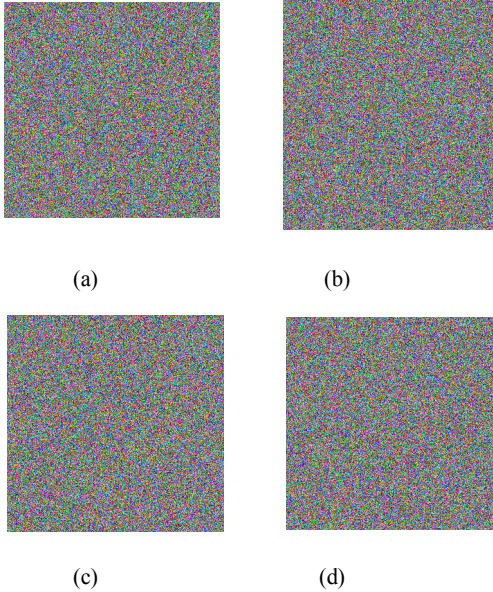


Fig. 6: Noise test using Gaussian noise (mean=0, variance=0.1), Poison (mean=10), Salt and Pepper (density= 0.05) and Speckle (mean=0, variance=0.04).

(vii) **Brute Force Resistance:** Brute force is a method of attacking keys of encryption algorithm by calculating all its

combinations. A small key length can be broken using this method but larger key length requires extensive computing resources and large time. In the proposed scheme using an externally generated key of length 128 bits is quite immune to brute force attack.

(viii) **Noise Test:** When transmitting the encrypted image over the network, cipher image may get affected with noise. In this test the cipher image is affected with noise using some known methods and their decryption is attempted. Figure: 6 (a-d) shows the corresponding decryption results. From the figure it is evident that the proposed algorithm successfully achieved the desired level of decryption. However, it fails when the noise in high volume of pixels is introduced.

(ix) **Crop Test:** The cipher image was cropped with a window of arbitrary size (M x N). The test results suggested that the deciphered image was majorly affected only in the area of cropping and less affected in other areas, which validates the proposed algorithm as per results depicted in Figure: 7.

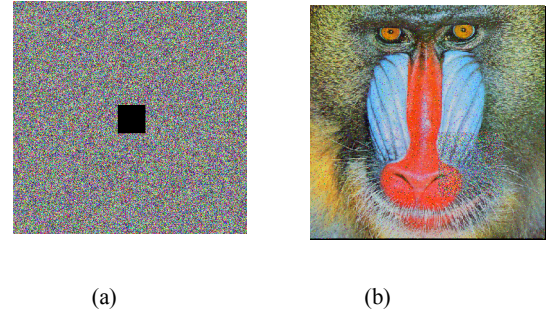


Fig.7. Crop test of the given image shows that the decrypted image contains some distortion spread in the entire image.

IX. CONCLUSION

The present work demonstrated a color image encryption using intertwining logistic map and elementary cellular automata. The encryption process consists of two key processes such as confusion and diffusion. The confusion process first permutes the pixels and then diffusion process propagates a minor change in a pixel to the entire image. As the algorithm is based on chaotic system, minor change in key results in entirely different outputs. Various tests discussed in the paper suggests that the system is able to resist brute force attack, differential attack, and works in error free transmission. On the basis of the results it can be concluded that the system is capable of encrypting digital color images for transmission over the network with higher security and reliability.

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REFERENCES

- [1] X. Huang, G. Ye, and K.-W. Wong, "Chaotic image encryption algorithm based on circulant operation," in *Abstract and Applied Analysis*, vol. 2013. Hindawi Publishing Corporation, 2013.
- [2] Y. Sun, L. Chen, R. Xu, and R. Kong, "An Image Encryption Algorithm Utilizing Julia Sets and Hilbert Curves," *PloS one*, vol. 9, no. 1, pp. 846–855, 2014.
- [3] M. Yang, N. Bourbakis, and S. Li, "Data-image-video encryption," *Potentials, IEEE*, vol. 23, no. 3, pp. 28–34, 2004.
- [4] E. Glasner and B. Weiss, "Sensitive dependence on initial conditions," *Nonlinearity*, vol. 6, no. 6, p. 1067, 1993.
- [5] G. Chen, Y. Mao, and C. K. Chui, "A symmetric image encryption scheme based on 3d chaotic cat maps," *Chaos, Solitons & Fractals*, vol. 21, no. 3, pp. 749–761, 2004.
- [6] J. Jin and Z.-h. Wu, "A secret image sharing based on neighborhood configurations of 2-d cellular automata," *Optics & Laser Technology*, vol. 44, no. 3, pp. 538–548, 2012.
- [7] A. Bakhshandeh and Z. Eslami, "An authenticated image encryption scheme based on chaotic maps and memory cellular automata," *Optics and Lasers in Engineering*, vol. 51, no. 6, pp. 665 – 673, 2013.
- [8] X. Wang and D. Xu, "Image encryption using genetic operators and intertwining logistic map," *Nonlinear Dynamics*, vol. 78, no. 4, pp. 2975–2984, 2014.
- [9] Y. Wu, J. P. Noonan, and S. Agaian, "NPCR and UACI randomness tests for image encryption," *Cyber Journals: Multidisciplinary Journals in Science and Technology, Journal of Selected Areas in Telecommunications (JSAT)*, pp. 31–38, 2011.
- [10] Y. Zhang, D. Xiao, Y. Shu, and J. Li, "A novel image encryption scheme based on a linear hyperbolic chaotic system of partial differential equations," *Signal Processing: Image Communication*, vol. 28, no. 3, pp. 292–300, 2013.
- [11] X. Tong, M. Cui, and Z. Wang, "A new feedback image encryption scheme based on perturbation with dynamical compound chaotic sequence cipher generator," *Optics Communications*, vol. 282, no. 14, pp. 2722–2728, 2009.
- [12] X.-y. Wang, F. Chen, and T. Wang, "A new compound mode of confusion and diffusion for block encryption of image based on chaos," *Communications in Nonlinear Science and Numerical Simulation*, vol. 15, no. 9, pp. 2479–2485, 2010.
- [13] M. Amin, O. S. Faragallah, and A. A. Abd El-Latif, "A chaotic block cipher algorithm for image cryptosystems," *Communications in Nonlinear Science and Numerical Simulation*, vol. 15, no. 11, pp. 3484–3497, 2010.
- [14] V. Patidar, G. Purohit, K. Sud, and N. Pareek, "Image encryption through a novel permutation-substitution scheme based on chaotic standard map," in *Chaos-Fractals Theories and Applications (IWCFTA)*, 2010 International Workshop on. IEEE, pp. 164–169, 2010.